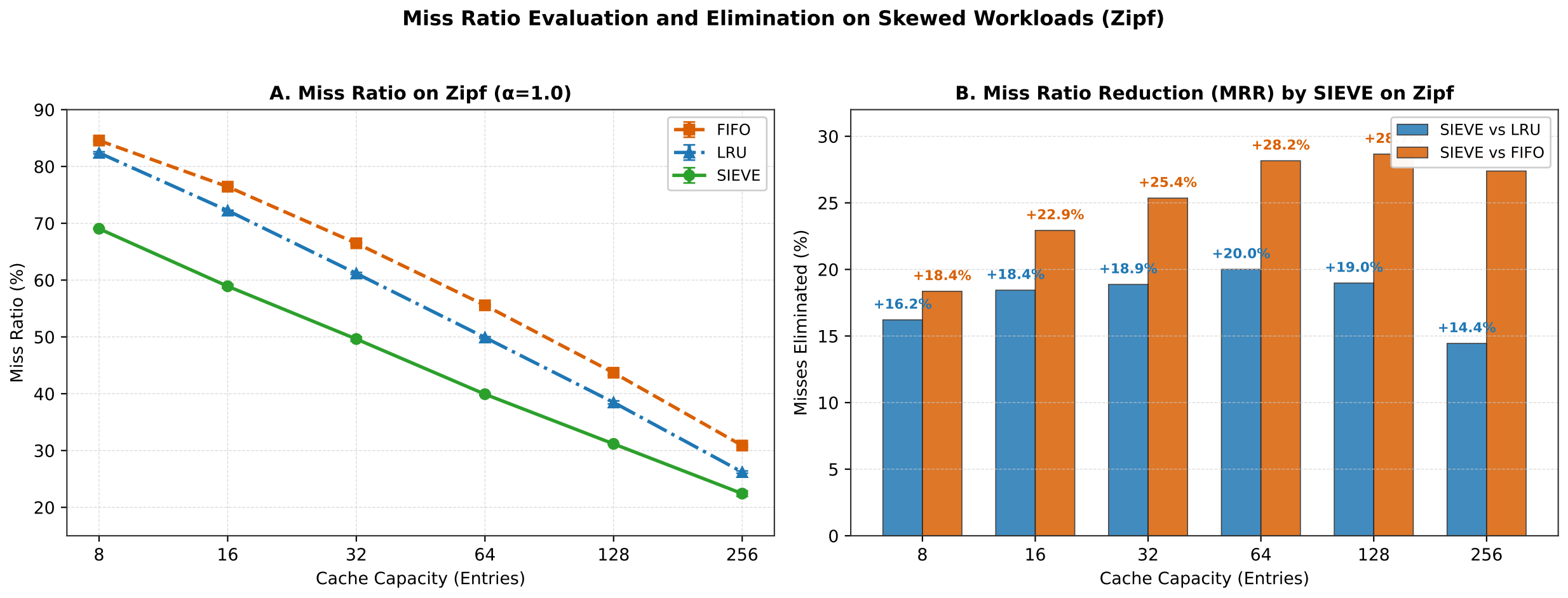
Miss Ratio Evaluation and Elimination on Skewed Workloads (Zipf)
A. Miss Ratio on Zipf (α=1.0)
Miss Ratio (%)
Cache Capacity (Entries)
FIFO
LRU
SIEVE
B. Miss Ratio Reduction (MRR) by SIEVE on Zipf
Misses Eliminated (%)
Cache Capacity (Entries)
SIEVE vs LRU
SIEVE vs FIFO
+16.2%
+18.4%
+18.4%
+22.9%
+18.9%
+25.4%
+20.0%
+28.2%
+19.0%
+14.4%